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Experimental investigation and thermodynamic modeling of the Ce-Si system

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ABSTRACT

Phase equilibria and thermodynamic properties of the Ce-Si system were studied experimentally by investigating a Si/(Ce,Si) diffusion couple and thirteen Ce-Si alloys. X-ray diffraction (XRD), electron probe micro-analyzer (EPMA), differential thermal analysis (DTA) and high temperature reaction calorimetry (HTRC) were utilized. A phase with close composition of $CeSi_{1.34}$ and the decomposition temperature of 1469 K is detected. The eutectic reaction liquid ↔ (Si) + $CeSi_2$ is observed to occurs at 1502.6 K. The enthalpy of formation of $CeSi_{1.68}$ was measured to be -70 ± 6 kJ mol-atoms^{-1}. A thermodynamic modeling of the Ce-Si system was then performed by considering the reliable literature data and the present experimental results. A substitutional solution model was used for the description of the liquid phase, and all of the compounds are treated as stoichiometric ones. The calculated phase diagram and thermodynamic properties using the obtained thermodynamic parameters are in good agreement with the experimental data.

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1. Introduction

Physical properties of the Ce-Si binary compounds and the higher order Ce-based systems have attracted a great deal of interests due to their electromagnetic properties and the potential in applications of semiconductor technology. In addition, it is reported that Ce can improve the tensile strength and ductility of Al-Si casting alloys through modifying their microstructure [1]. Alloy development requires an understanding of the phase equilibria and thus the thermodynamic description for the Ce-Si system is highly needed.

The Ce-Si system was previously thermodynamically assessed by Gröbner et al. [2], Shukla et al. [3] and Kim et al. [4]. Using a substitutional solution model, Gröbner et al. [2] assessed the phase diagram mainly based on the melting points and the standard enthalpies of formation of the compounds measured by Benesovsky et al. [5] and Meschel et al. [6,7] respectively. However, their calculated mixing enthalpies of the liquid phase showed a large deviation from the experimental data [8,9]. Later, Shukla et al. [3] performed a thermodynamic optimization for this system using a Modified Quasi-chemical Model (MQM) to describe the short-range ordering of the liquid phase. More experimental data were taken into consideration except for the heat capacities at the low temperature and heat content of Ce_5Si_3 measured by Gorbachuk [10] since the calculated entropy of Ce_5Si_3 at 298 K based on their optimization was about 60 J·K^{-1}·mol-atoms^{-1} higher than the experimental data [10]. Besides, their assumption that the heat capacities of those compounds from Ce_5Si_3 to CeSi vary linearly with atomic fraction is not necessary. Most recently, a better described adopted MQM was given by Kim et al. [4], who improved the issue in the work of Shukla et al. [3]. However, as Kim et al. [4] pointed out there are discrepancies in the homogeneity range and the polymorphs of $CeSi_{2-a}$ [5,11–15]. In addition, when looking closely at the literature data [13,16], the reported phase transition temperatures of Liquid + (Si) + $CeSi_2$ are about 50 K from each other. Moreover, one possible compound with the composition of $CeSi_{1.34}$ was reported in literature [17] but not considered in the previous assessments [2–4]. All these need further experimental clarification. This present work aims to clarify the phase diagram experimentally and obtain a reliable thermodynamic description of the Ce-Si binary system by CALPHAD methodology. The work is also part of the project to develop a multi-component Al-base thermodynamic database [18].

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Table 1
Summary of the experimental data of the Ce-Si system.

Ref.	Experimental technique[b]	Type of data	QM[a]
[16]	TA	Liquidus, 68–100 at.% Si eutectic composition and temperature of the reaction Liquid ↔ (Si) + $CeSi_2$	– +
[5]	XRD, MM	Melting points of the compounds Eutectic temperature of the reaction Liquid ↔ (Si) + $CeSi_2$	– –
[13]	DTA, XRD, MM	Phase diagram in all range	+
[14,15]	EPMA, XRD, DTA	Phase diagram, 60–70 at.% Si	+
[6,7]	HTRC	enthalpies of formation of CeSi, $CeSi_2$ and Ce_5Si_3	+
[8,9]	IC	1923 K, enthalpy of mixing in the liquid	+
[23]	IC	1890 K, enthalpy of mixing in the liquid	+
[24]	SHPM	up to 25 K, heat capacity of Ce_3Si_2	–
[10]	AM, HDC	55 K to 298 K, heat capacity of Ce_5Si_3 437 to 1806 K, heat contents of Ce_5Si_3	+ +
TW[c]	XRD, DTA, EPMA	Temperature for eutectic reaction Liquid ↔ (Si) + $CeSi_2$	+
TW[c]	XRD, DTA, HTRC	enthalpies of formation of $CeSi_{1.68}$	+

[a] QM means Quoted Mode, which indicates whether the data are used in the parameter optimization: + used; – not used.
[b] TA, thermal analysis; XRD, X-ray diffraction; MM, micrographic method; DTA, differential thermal analysis; EPMA, electron probe microanalysis; HTRC, high temperature reaction calorimeter; IC, isothermal calorimetry; SHPM, semi-adiabatic heat pulse method; AM, adiabatic method; HDC, high-temperature differential calorimeter.
[c] TW, this work.

Table 2
Crystallographic data on the binary compounds in the Ce-Si system from literature [13,17].

Phase	Prototype	Pearson symbol	Space group	Refs./Comments
Ce_5Si_3	Cr_5B_3	*tI32*	I4/mcm	[13]
Ce_3Si_2	U_3Si_2	*tP10*	P4/mbm	[13]
Ce_5Si_4	Zr_5Si_4	*tP36*	$P4_12_12$	[13]
CeSi	FeB	*oP8*	Pnma	[13]
$CeSi_{1.34}$	V_2B_3	*oS20*	Cmcm	[17]
$CeSi_{1.68}$	$\alpha GdSi_2$	*oI12*	Imma	[13]
$CeSi_2$	$\alpha ThSi_2$	*tI12*	$I4_1/amd$	[13]

2. Literature review

The Ce-Si system has been reviewed by Munitz et al. [19] based on the available experimental phase diagram and thermodynamic data up to 1989 [5,16]. The present assessment takes into account both the experimental data assessed by Munitz et al. [19] and those published later [6,7,10,13–15,17]. In order to facilitate reading, the experimental information of the Ce-Si system is briefly summarized with the quoted mode for each reference in Table 1. All the crystal data relevant to the phases in the Ce-Si system are compiled in Table 2.

2.1. Phase diagram data

Vogel [16] investigated the liquidus from 68 to 100 at.% Si, and detected that a eutectic reaction involving silicon that occurs at 81.5 at.% Si and 1513 K. There was at least one compound [16], probably of the formula CeSi. Using X-ray powder diffraction (XRD), metallographic analyses and melting point measurements, Benesovsky et al. [5] reported the existence of six intermediate phases. $CeSi_{2\text{-}a}$ was reported to melt incongruently while the other five silicides, $Ce_{\sim 2}Si$, Ce_3Si_2, $Ce_{\sim 1.2}Si$, CeSi and $CeSi_2$ melt congruently at 1673, 1663, 1713, 1743 and 1893 K, respectively. The eutectic reaction involving silicon reported by Vogel [16] was determined to be Liquid ↔ (Si) + $CeSi_2$ with a temperature of 1463 K, which is 50 K lower than that reported by Vogel [16]. Later, $Ce_{\sim 2}Si$ and $Ce_{\sim 1.2}Si$ was amended to be Ce_5Si_3 [20] and Ce_5Si_4 [21],respectively. Bulanova et al. [13] re-investigated the Ce-Si phase diagram by means of differential thermal analysis (DTA), XRD and metallography examination. Ce_5Si_3, Ce_3Si_2, Ce_5Si_4, CeSi, $CeSi_{1.68}$ and $CeSi_2$ were verified. Ce_5Si_3, Ce_3Si_2, Ce_5Si_4 and $CeSi_{1.68}$ were reported to be formed peritectically, which are different from the conclusions given by Benesovsky et al. [5]. Most recently, the phase diagram reported by Bulanova et al. [13] was slightly modified by Souptel et al. [14,15], who prepared $CeSi_x$ (x = 2.0, 1.89, 1.86, 1.84, 1.82, 1.81, 1.80, 1.76) single crystals by a floating zone technique and performed electron probe microanalysis (EPMA), XRD and DTA. The congruent melting of $CeSi_{1.68}$ was confirmed, while the eutectic transformation of Liquid ↔ CeSi + $CeSi_{1.68}$ occurs at 1887 K, which is 81.5 K higher than that reported by Bulanova et al. [13].

As for the phase relationship in the range of 60–70 at.% Si, there are many discrepancies in the literature. Benesovsky et al. [5] reported two intermediate phases $CeSi_{2\text{-}a}$ and $CeSi_2$ crystallized with the same $\alpha ThSi_2$-type structure. However, crystallographic data later published indicated that there are two structures: $\alpha GdSi_2$-type on the Si-poor side and $\alpha ThSi_2$-type on the Si-rich side. Byanov et al. [11] believed there is only one phase with a high-temperature $\alpha ThSi_2$ structure and a low-temperature $\alpha GdSi_2$ structure, while Houssay et al. [12] proposed that a continuous transition from $\alpha GdSi_2$-type to $\alpha ThSi_2$-type occurs when the Si content increases from 61.5 to 67 at.%. Subsequently, the presence of two intermediate phases $CeSi_{1.68}$ and $CeSi_2$ with different structures was reported by Bulanova et al. [13] and supported by the work of Souptel et al. [14,15]. However, the exact homogeneity ranges of these two phases are not clear. Yashima et al. [22] reported the compound $CeSi_{1.68}$ containing 63 at.% Si. However, Munitz et al. [19] reported that the homogeneity region of $CeSi_2$ is from 64 to 66.7 at.% Si while $CeSi_{1.68}$ shows no solubility with composition of 62.5 at.% Si. The study of Bulanova et al. [13] indicated that the boundaries of the two-phase region of $CeSi_{2\text{-}a1}$ + $CeSi_{2\text{-}a2}$ are located between 62.6 and 64.1 at.% Si and shift towards Si-poorer compositions when the temperature decreases. In correspondence with Houssay et al. [12], the results of Bulanova et al. [13] indicated the lowest Si content in $CeSi_{1.68}$ and the highest in $CeSi_2$ is 61.5 and 66.7 at.% Si, respectively. As regards the other compounds in the Ce-Si system, no observable homogeneity range was reported.

In addition to the six intermediate phases reported above, a new compound $Ce_2Si_{3\text{-}y}$ with y = 0.3 (or $CeSi_{1.34}$) was reported by Schobinger-Papamantellos et al. [17] by means of neutron diffraction and magnetic measurements, but the existence of this compound was not mentioned in the work of Bulanova et al. [13].

2.2. Thermodynamic data

The standard enthalpies of formation of CeSi, $CeSi_2$ and Ce_5Si_3 were determined by Meschel et al. [6,7] using high temperature reaction calorimetry (HTRC). The integral enthalpy of mixing in the liquid phase was measured by Ryss et al. [8,9] at 1923 K and Sudavtsova et al. [23] at 1890 K using isothermal calorimetry. The lowest integral enthalpy (-74 kJ mol-atoms^{-1}) of formation was determined at the composition of 52 at.% Si by Ryss et al. [8,9].

Additionally, the heat capacity of Ce_3Si_2 at low temperature was determined by Dhar et al. [24]. The heat capacity from 55 K to 298 K and the enthalpy increments (H_T—$H_{298.15K}$) over a wide temperature range of Ce_5Si_3 were studied by Gorbachuk [10] by means of

Table 3
Sources and purities of the chemicals used in the study.

Chemical	Source	Purity (wt.%)	Purification method
Ce	Alfa Aesar	99.9	None
Si	Alfa Aesar	99.999	None

adiabatic method and high-temperature differential calorimeter, respectively.

3. Experimental investigation

The samples were prepared from high purity metals Ce (99.9 wt.%) and Si (99.999 wt.%). The source and the purity of the starting chemicals are summarized in Table 3.

A solid–liquid contact method was employed to produce a Si/(Ce,Si) diffusion couple. The diffusion couple was assembled by putting a Si slice and a pre-prepared Ce-8 at.% Si alloy block together into a titanium crucible and sealed in a vacuum quartz tube under high vacuum of 1 Pa. The quartz capsule was then annealed at 1373 K for three hours and subsequently quenched in cold water. The quenched diffusion couple was mounted carefully to reveal the exact vertical interface.

Thirteen Ce–Si binary alloys were prepared by arc melting the weighted pure Ce pieces and Si chips in an arc furnace (WKDHLI, Beijing Opto-electronics Co. Ltd., China) under high purity argon atmosphere of 101325 Pa using a non-consumable tungsten electrode. The alloy ingots were re-melted at least six times to improve homogeneity. The mass losses were less than 0.5 wt.% and therefore the samples were not subjected to chemical analysis. The as-cast alloys were wrapped in tungsten foils and encapsulated in quartz capsules under high vacuum of 1 Pa and annealed in an L4514-type diffusion furnace (Qingdao Instrument & Equipment Co Ltd., China) at 1373 K for 26 days, followed by quenching in cold water.

The diffusion couple and key alloys were characterized by using optical microscopy and then analyzed with EPMA (JXA-8530, JEOL, Japan) employing pure Ce (99.99 wt.%) and Si (99.9 wt.%) as standard materials. The EPMA measurements were analyzed at 15 kV and 2×10^{-8} A.

XRD measurements of the annealed alloys were conducted using a Cu-K_α radiation ($\lambda = 0.15405$ nm) on a Rigaku D-max/2550 VB+ X-ray diffractometer at 40 kV and 250 mA with a scan step of 0.02° over a 2θ range of 10–85°. High purity Ge was added as the internal standard for calibrating the XRD patterns of the representative samples.

Alloys #1 ($Ce_{0.08}Si_{0.92}$) and #2 ($Ce_{0.23}Si_{0.77}$) were subject to DTA (DSC404C, Netzsch, Germany) measurement. The sample holder material for DTA measurement was pure Al_2O_3. In the light of the estimated melting points, measurements were performed in the temperature ranges of 1373–1673 K and 1373–1773 K, respectively, with heating and cooling rates of 10 K/min in a flowing argon atmosphere. Since the conventional open crucible may not be suitable for measuring those samples with relatively high Ce content due to the high volatilization of Ce at elevated temperature, alloys #12 ($Ce_{0.45}Si_{0.55}$) and #13 ($Ce_{0.47}Si_{0.53}$) were subjected to differential scanning calorimetry (DSC) (DSC404C, NETZSCH, Germany) after being sealed into Ta crucibles in a glove box using arc melting. The measurements of alloys #12 and #13 were performed from 1373 K to 1723 K with a heating and cooling rate of 10 K/min. Before the analysis, the calibration was performed by measuring the melting temperatures of In (429.75 K), Sn (505.05 K), Bi (544.55 K), Al (933.45 K), Ag (1234.95 K) and Au (1337.35 K) for both DTA and DSC, as well as Ni (1728.30 K) for DTA. The S type thermocouple was used and the accuracy of the temperature measurement was estimated to be ± 2 K. Temperatures of invariant reactions and the liquidus were recorded from the onset of the first peak and the maximum of the second peak on the heating curves, respectively.

Table 4
The tie-line data (EPMA) from the Si/(Ce,Si) diffusion couple at 1373 K for 3 h (composition in at.% Si).

phase 1	Composition[a]	phase 2	Composition[a]
(Si)	end-member	$CeSi_2$	67.0
$CeSi_2$	64.2	$CeSi_{1.68}$	63.8
$CeSi_{1.68}$	63.2	$CeSi_{1.34}$	58.0
$CeSi_{1.34}$	57.4	CeSi	51.2
CeSi	49.9	Ce_5Si_4	44.9
Ce_5Si_4	45.1	Ce_3Si_2	40.2
Ce_3Si_2	40.3	Ce_5Si_3	38.2
Ce_5Si_3	–	Ce_5Si_3 + (Ce)	end-member

[a] EPMA investigations were carried out at T = 293 K and in argon gas of $P = 4 * 10^{-4}$ Pa. Standard uncertainties u are: u(T) = 1.0 K, u(P) = 0.5%. The concentrations were detected in both sides of 1–2 μm from the phase boundaries. Standard uncertainty for concentration is u(x) = 0.1 at%.

In order to determine the enthalpy of formation of $CeSi_{1.68}$, a Kleppa-type High Temperature Reaction Calorimeter (HTRC) was utilized with the calorimeter temperature set at 1373 ± 2 K. The experiment was carried out under the protection of argon gas. Six samples, each with a weight of about 0.1 g, were produced by mixing powder of Ce and Si in a mortar according to the molar ratio of 1:1.68, and pressed into small pellets then dropped into the calorimeter from 298 K. The reaction product at 1373 K was dropped from 298 K again into the calorimeter to obtain the heat content of the compound. The samples were examined by XRD to assess their structures and to ascertain the absence of unreacted metals.

The standard enthalpies of formation of $CeSi_{1.68}$ determined in this work was obtained from the difference between the results of the two sets of measurements.

$$Ce(s, 298K) + 1.68^{*}Si(s, 298K) \leftrightarrow CeSi_{1.68}(s, 1373K),\ \Delta H_1 \quad (1)$$

$$CeSi_{1.68}(s, 298K) \leftrightarrow CeSi_{1.68}(s, 1373K),\ \Delta H_2 \quad (2)$$

The standard enthalpy of formation is given by the expression as below:

$$\Delta_f H^\circ = \Delta H_1 - \Delta H_2 \quad (3)$$

4. Thermodynamic modeling

4.1. Pure elements

The Gibbs free energies G_i^φ for the element i (i = Ce, Si) in the phase φ are taken from SGTE Data for Pure Elements compiled by Dinsdale [25] and described with the following equation:

$$G_i^\varphi - H_i^{SER} = a + bT + cTlnT + dT^2 + eT^{-1} + fT^3 + gT^7 + hT^{-9} \quad (4)$$

where H_i^{SER} is the molar enthalpy of the element i at 298.15 K and 1 bar in its standard element reference (SER) state, face-centered cubic (fcc) for Ce and diamond for Si, and T is the absolute temperature. The last two terms in Eq. (4) are used only outside the ranges of the melting point, gT^7 for a liquid below the melting point and hT^{-9} for solid phases above the melting point.

4.2. Stoichiometric intermetallic compounds

Since no obvious solubility ranges were reported, all the intermetallic compounds were treated as stoichiometric compounds. The Gibbs energies per mole formula of the compounds except Ce_5Si_3 are described by the following expression:

$$^0G_m^{Ce_xSi_y} - xH_{Ce}^{SER} - yH_{Si}^{SER} = a + bT + x^0G_{Ce}^{fcc} + y^0G_{Si}^{diamond} \quad (5)$$

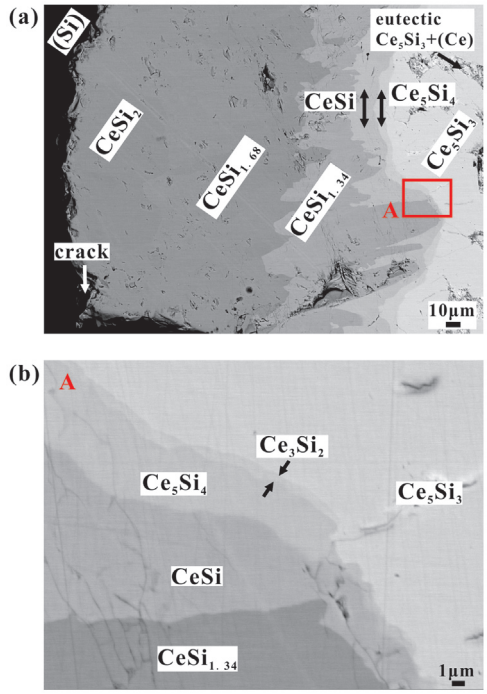

Fig. 1. Microstructure of the Ce-Si diffusion couple annealed at 1373 K for 3 h. (a) full view; (b) enlargement of region A.

where 'a' and 'b' are variables to be evaluated in the assessment.

For Ce_5Si_3, there are experimental data on heat capacity (C_p) around 298 K and enthalpy increments. It is preferable to express its Gibbs energy relative to the SER state. The Gibbs energy of Ce_5Si_3 per mol-formula is expressed as:

$$G_m^{Ce_5Si_3} - 5H_{Ce}^{SER} - 3H_{Si}^{SER} = a + bT + cTlnT + dT^2 + eT^{-1} \quad (6)$$

The coefficients *c*, *d* and *e* can be obtained directly from the C_p expression.

4.3. The liquid phase

Liquid phase is treated as a substitutional solution by the random solution model in the form of (Ce,Si) with the Gibbs energies expressed as:

$$G_m^{liquid} = x_{Ce}^0 G_{Ce}^{liquid} + x_{Si}^0 G_{Si}^{liquid} + RT(x_{Ce} lnx_{Ce} + x_{Si} lnx_{Si}) + {}^{ex}G_m^{liquid} \quad (7)$$

${}^{ex}G_m^{liquid}$ is the excess Gibbs energy, expressed in Redlich–Kister polynomial as follows:

$${}^{ex}G_m^{liquid} = x_{Ce}x_{Si}\sum_{j=0}^{n} {}^{j}L_{Ce,Si}^{liquid}(x_{Ce} - x_{Si})^j \quad (8)$$

where ${}^{j}L_{Ce,Si}^{liquid}$ is the *j* th-order binary interaction parameter expressed as ${}^{j}a + {}^{j}bT$, and '${}^{j}a$' and '${}^{j}b$' are model parameters to be evaluated from experimental information.

5. Results and discussion

5.1. Experimental results

The data on the tie–lines from the annealed Si/(Ce,Si) diffusion couple determined using EPMA are listed in Table 4. Table 5 gives the nominal compositions of the binary alloys, the phases identified by a combination of XRD and EPMA, as well as the phase

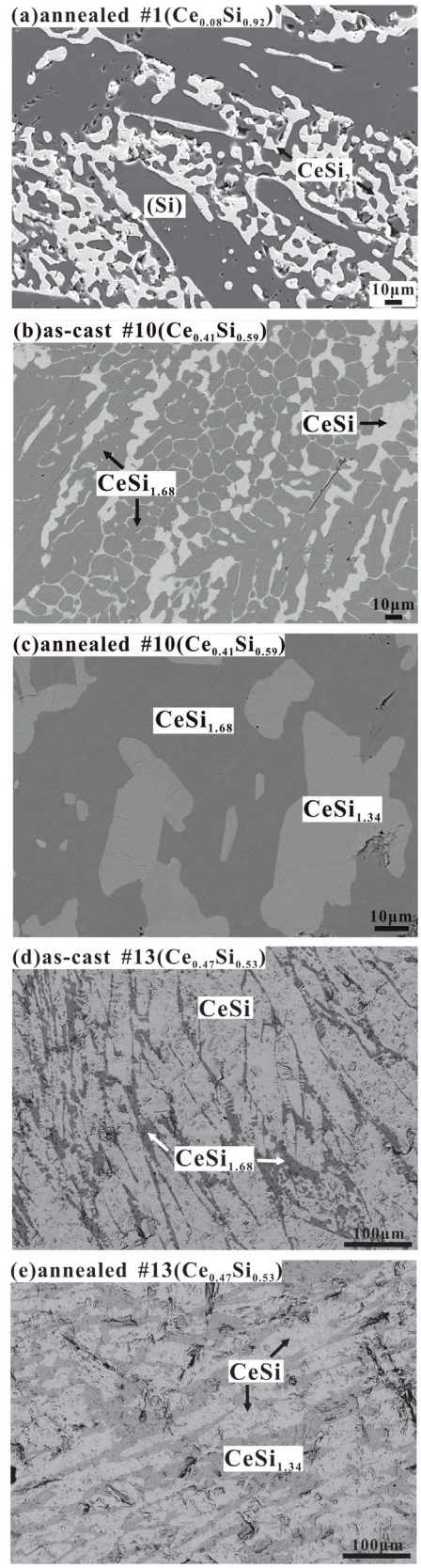

Fig. 2. Microstructures of the representative alloys. (a) annealed alloy #1 ($Ce_{0.08}Si_{0.92}$); (b) as-cast alloy #10 ($Ce_{0.41}Si_{0.59}$); (c) annealed alloy #10 ($Ce_{0.41}Si_{0.59}$); (d) as-cast alloy #13 ($Ce_{0.47}Si_{0.53}$); (e) annealed alloy #13 ($Ce_{0.47}Si_{0.53}$).

Table 5
Summary of the XRD and EPMA results of the alloys annealed at 1373 K for 26 days (composition in at.% Si, data from as-cast alloys are marked by *).

No.	Nominal Composition	Phase	Lattice parameters[a], Å			Composition[b]
			a	*b*	*c*	
1	92	(Si)	–	–	–	–
		$CeSi_2$	4.191 ± 0.003	–	13.947 ± 0.001	–
2	77	(Si)	–	–	–	99.9
		$CeSi_2$	4.192 ± 0.001	–	13.961 ± 0.001	66.6
3	68	(Si)	–	–	–	99.8
		$CeSi_2$	4.194 ± 0.001	–	13.942 ± 0.001	66.8
4	67	$CeSi_2$	4.195 ± 0.001	–	13.934 ± 0.001	–
5	66	$CeSi_2$	4.180 ± 0.001	–	13.844 ± 0.001	–
6	65	$CeSi_2$	4.193 ± 0.001	–	13.882 ± 0.001	–
7	64	$CeSi_2$	4.188 ± 0.001	–	13.873 ± 0.001	–
8	63.5	$CeSi_{1.68}$	–	–	–	63.6
9	63	$CeSi_{1.68}$	4.125 ± 0.001	4.191 ± 0.001	13.878 ± 0.001	–
10	59	$CeSi_{1.68}$	–	–	–	63.1
		$CeSi_{1.34}$	–	–	–	56.9
11	57	$CeSi_{1.34}$	–	–	–	56.8
		CeSi (trace)	–	–	–	49.5
12	55	$CeSi_{1.34}$		–	–	57.3
		CeSi		–	–	50.5
		$CeSi_{1.68}$*	–	–	–	63.6
		CeSi*	–	–	–	50.1
13	53	$CeSi_{1.34}$	–	–	–	57.6
		CeSi	–	–	–	50.8
		$CeSi_{1.68}$*	–	–	–	64.0
		CeSi*	–	–	–	50.3

[a] XRD investigations were carried out at T = 298 K and in air of P = 101325 Pa. Standard uncertainties u are: u(T) = 1.0 K, u(P) = 5000 Pa. Uncertainties for lattice parameters are expressed as standard uncertainty.
[b] EPMA investigations were carried out at T = 293 K and in argon gas of P = 4 * 10^{-4} Pa. Standard uncertainties u are: u(T) = 1.0 K, u(P) = 0.5%. Standard uncertainty for concentration is u(x) = 0.1 at%.

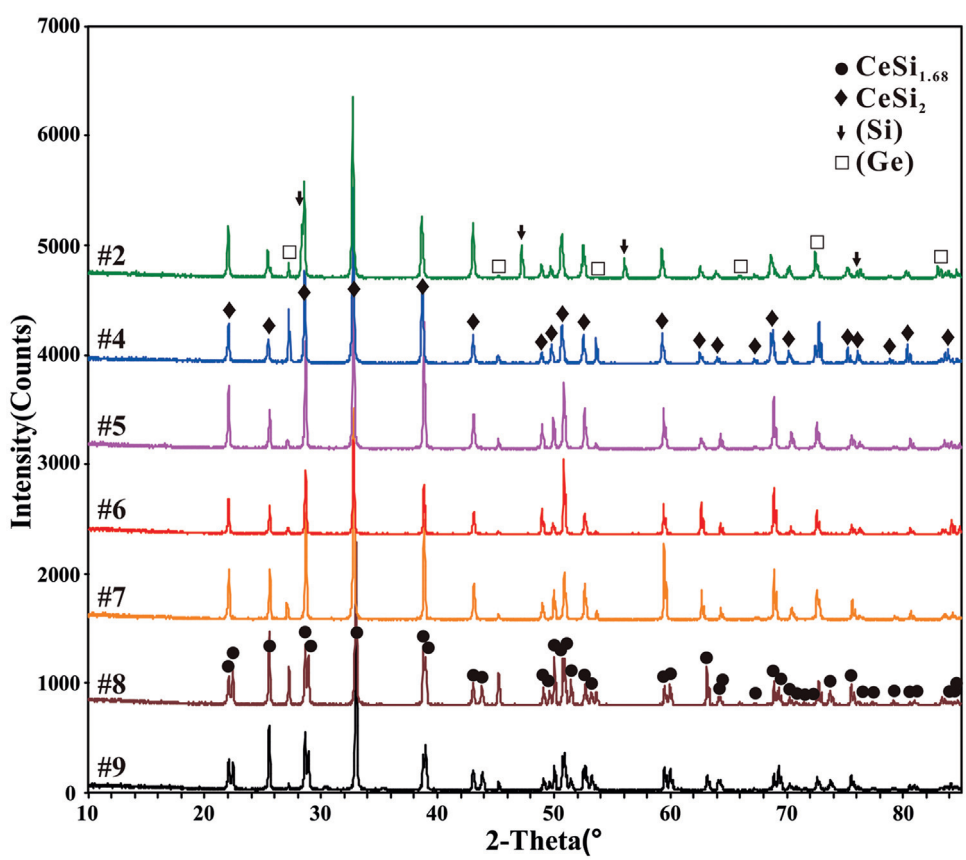

Fig. 3. XRD patterns of the alloys #2 ($Ce_{0.23}Si_{0.77}$), #4 ($Ce_{0.33}Si_{0.67}$), #5 ($Ce_{0.34}Si_{0.66}$), #6 ($Ce_{0.35}Si_{0.65}$), #7 ($Ce_{0.36}Si_{0.64}$), #8 ($Ce_{0.365}Si_{0.635}$) and #9 ($Ce_{0.37}Si_{0.63}$) annealed at 1373 K for 26 days.

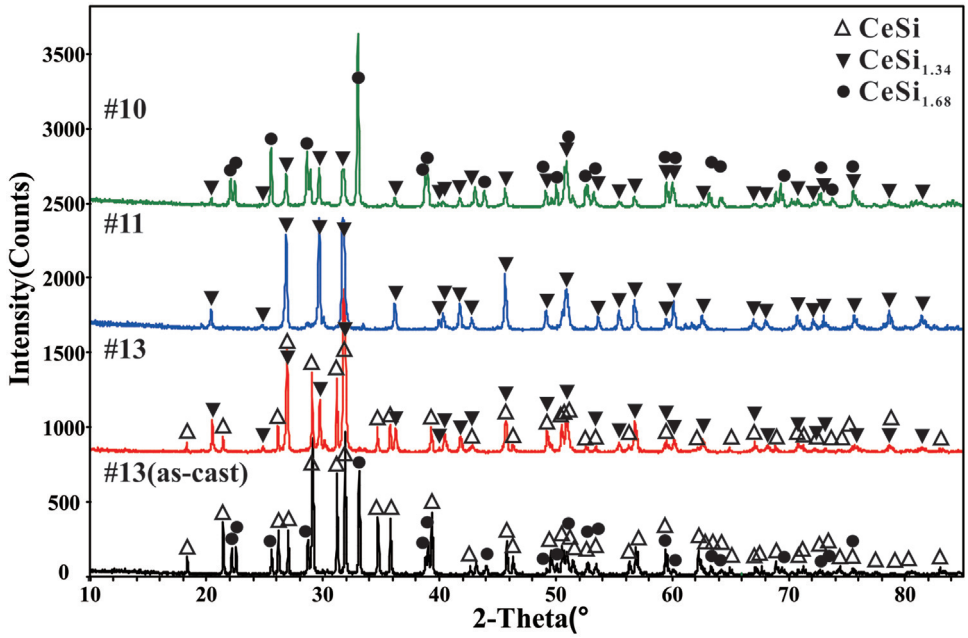

Fig. 4. XRD patterns of the alloys #10 ($Ce_{0.41}Si_{0.59}$), #11 ($Ce_{0.33}Si_{0.57}$) and #13 ($Ce_{0.47}Si_{0.53}$) annealed at 1373 K for 26 days comparing with the one from as-cast #13 ($Ce_{0.47}Si_{0.53}$).

compositions determined by EPMA. The microstructure of the diffusion couple is shown in Fig. 1. The backscattered electron (BSE) images and XRD patterns of the representative alloys are presented in Figs. 2–4 .

5.1.1. Diffusion couple

Fig. 1 shows the microstructure of the Si/(Ce,Si) solid–liquid diffusion couple DC #1 annealed at 1323 K for 3 h. Seven interdiffusion layers are visible between the pure Si on the left and the eutectic structure of (Ce) + Ce_5Si_3 on the right. All the six well accepted phases in the Ce-Si system, namely $CeSi_2$, $CeSi_{1.68}$, CeSi, Ce_5Si_4, Ce_3Si_2 and Ce_5Si_3, were identified in DC #1 by EPMA. It should be noted that there is an additional layer with about 57.4–58.0 at.% Si between the layers of $CeSi_{1.68}$ and CeSi, which has a proximate composition with the $CeSi_{1.34}$ phase reported by Schobinger-Papamantellos et al. [17]. Here, we call this phase as $CeSi_{1.34}$ considering its composition. The length of the Ce_3Si_2 interdiffusion layer is merely around 1 μm and can only be observed under high magnification, as indicated in Fig. 1(b), while other layers are in the range from 10 to 90 μm. It is worth noting that $CeSi_2$ and $CeSi_{1.68}$ have close compositions and show small contrast. The Si content in the layers of $CeSi_2$ and CeSi was determined to be 67.0–64.2 and 49.9–51.2 at.%, respectively.

5.1.2. Alloys

Fig. 2(a) shows the microstructure of the annealed alloy #1 ($Ce_{0.08}Si_{0.92}$). Combining with the XRD result in Fig. 3(a), two phases were identified to coexist: the dark solid solution (Si) and the light $CeSi_2$ phase. The same phase relationship can also be observed in the annealed alloys #2 ($Ce_{0.23}Si_{0.77}$) and #3 ($Ce_{0.32}Si_{0.68}$). The maximum Si content in $CeSi_2$ is determined to be 66.8 at.%, which agree well with the result from the diffusion couple aforementioned.

Alloys #4–9 were designed to study the phase relationship between $CeSi_{1.68}$ and $CeSi_2$. However, it was not possible to determine the phase boundaries [13] of these two phases because of the small contrast in the BSE images and the similarity in the XRD patterns between these two phases. As shown in Fig. 3, the XRD results indicate that the alloys #4 ($Ce_{0.33}Si_{0.67}$), #5 ($Ce_{0.34}Si_{0.66}$), #6 ($Ce_{0.35}Si_{0.65}$) and #7 ($Ce_{0.36}Si_{0.64}$) are located in the $CeSi_2$ single-phase region with the $\alpha ThSi_2$ structure, while the alloys #8 ($Ce_{0.365}Si_{0.635}$) and #9 ($Ce_{0.37}Si_{0.63}$) are located in the $CeSi_{1.68}$ ($\alpha GdSi_2$ structure) single-phase region.

Alloys #10–13 were prepared for investigating the existence of $CeSi_{1.34}$. Figs. 2(b)–(e) present the microstructures of the alloys #10 ($Ce_{0.41}Si_{0.59}$) and #13 ($Ce_{0.47}Si_{0.53}$) in both as-cast and annealed states. As shown in Fig. 2s(b) and (d), the eutectic structure of $CeSi_{1.68}$ + CeSi is observed in the two as-cast alloys. After annealing at 1373 K for 26 days, the eutectic structure in both alloys vanishes. A two-phase equilibrium between $CeSi_{1.68}$ and another phase with the composition close to $CeSi_{1.34}$ is detected in the alloy #10 while CeSi and the $CeSi_{1.34}$ phase are observed in the alloy #13, which is consistent with the observation from the aforementioned diffusion couple. The two-phase equilibrium of CeSi + $CeSi_{1.34}$ was also characterized in the annealed alloys #11 ($Ce_{0.43}Si_{0.57}$) and #12 ($Ce_{0.47}Si_{0.55}$) with different phase amounts. Fig. 4 compares the XRD patterns of both as-cast and annealed alloy #13 ($Ce_{0.47}Si_{0.53}$) with the annealed alloys #11 ($Ce_{0.43}Si_{0.57}$) and #10 ($Ce_{0.41}Si_{0.59}$). The unknown peaks marked by solid inverted triangles in Fig. 4 are detected in all of the three annealed alloys but cannot be indexed with any already accepted phases in this system, nor even fully by the one reported in Schobinger-Papamantellos et al. [17] with the similar composition as $CeSi_{1.34}$. Those reflections are assigned to a new phase and we still call it $CeSi_{1.34}$ based on its composition by EPMA. It should be noticed that although a trace amount of CeSi can be detected by EPMA in the annealed #11 ($Ce_{0.43}Si_{0.57}$), as indicated in Table 5, the XRD patterns of this alloy show only the presence of $CeSi_{1.34}$, with no visible amount of the CeSi. Besides, the minimum Si content in $CeSi_{1.68}$ is detected to be 63.1 at.%, which is higher than the datum reported in Munitz et al. [19] and Bulanova et al. [13] but agree well with that from Yashima et al. [22].

5.1.3. Thermodynamic properties

The DTA curves of alloys #1 ($Ce_{0.08}Si_{0.92}$) and #2 ($Ce_{0.23}Si_{0.77}$) are shown in Fig. 5, where two reaction peaks along the heating curves were detected in both alloys. The first peak with an onset at 1501.5 K from alloy #1 ($Ce_{0.08}Si_{0.92}$) and 1503.7 K from alloy #2 ($Ce_{0.23}Si_{0.77}$) represent the eutectic reaction liquid 0 ↔ (Si) + $CeSi_2$, which is higher than the reported temperatures by Benesovsky et al. [5] and Bulanova et al. [13] but close to the result given by Vogel [16]. The temperature of 1629.0 K from alloy #1 ($Ce_{0.08}Si_{0.92}$) and

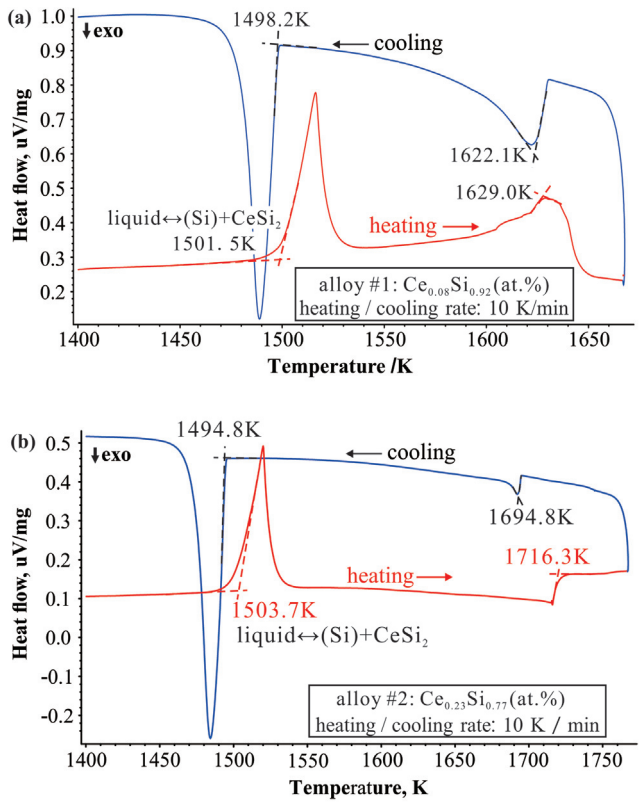

Fig. 5. DTA trace with a heating and cooling rate of 10 K/min of the annealed alloys (a) #1 ($Ce_{0.08}Si_{0.92}$); (b) #2 ($Ce_{0.23}Si_{0.77}$).

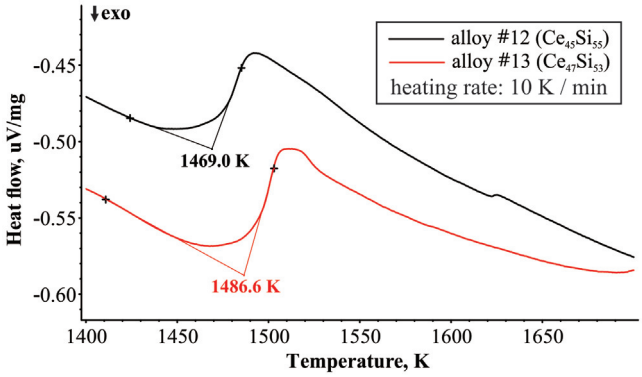

Fig. 6. DSC curves with a heating rate of 10 K/min of the annealed alloys #12 ($Ce_{0.45}Si_{0.55}$) and #13 ($Ce_{0.47}Si_{0.53}$).

1716.3 K from alloy #2 ($Ce_{0.23}Si_{0.77}$) are assumed as the melting points of the corresponding alloys.

Fig. 6 shows the DSC curves during heating for the annealed alloys #12 ($Ce_{0.45}Si_{0.55}$) and #13 ($Ce_{0.47}Si_{0.53}$). An endothermic peak, which may be associated with the peritectoid reaction $CeSi + CeSi_{1.68} \leftrightarrow CeSi_{1.34}$, is detected in the two alloys with a temperature of 1469.0 and 1486.6 K, respectively. Whereas on cooling, no endothermic or exothermic events were recorded and therefore the cooling curves are not presented in the figure. The deviation in the detected temperature and the absence of thermal signals on cooling is perhaps due to the sluggishness of this peritectoid reaction.

The enthalpy of formation of $CeSi_{1.68}$ was measured to be -70 ± 6 kJ mol-atoms^{-1} by HTRC. It was quite difficult to prepare single phase samples but the minor amount of other phases would not introduce noticeable errors to the enthalpy value [26]. Hence,

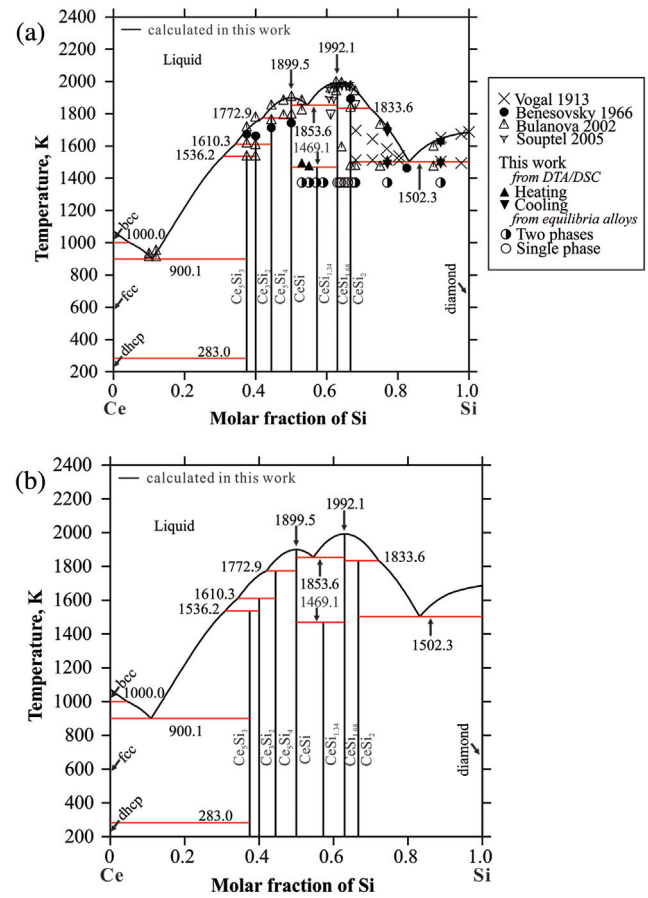

Fig. 7. Calculated Ce-Si phase diagram based on this optimization (a) along with the experimental data from present work and the literature [5,13–16]; (b) without experimental data.

the measured enthalpy of formation for $CeSi_{1.68}$ was considered to be reliable.

5.2. Thermodynamic assessment

Evaluation of the model parameters was attained by recurrent run of the embed PARROT module in the Thermo-Calc software [27], which works by minimizing the sum of squares of the differences between measured and calculated data. The thermodynamic parameters obtained in the present work are listed in Table 6.

Using the present set of the thermodynamic parameters, the calculated Ce-Si phase diagram, shown in Fig. 7, agrees well with the experimental data from the literature [5,13–16] and the present experimental results. Table 7 compares all the calculated and measured invariant reaction temperatures. For the invariant reaction liquid $\leftrightarrow$ CeSi + $CeSi_{1.68}$, there is a large deviation of 81.5 K between the only two sets of experimental data in literature [13–15]. Considering the available thermodynamic properties, it looks hard to reproduce any of two data without causing significant deviation in the related phase relations or thermodynamic properties of liquid. Therefore, none of these two data was used in this work.

Fig. 8 shows a comparison of the calculated enthalpies of mixing of liquid with the experimental data [8,9,23]. Since the error bars of the work by Ryss et al. [8,9] were not given, and one should notice their experimental temperature (1923 K) is lower than the melting temperature of $CeSi_{1.68}$ (1993 K) and across the two-phase regions, the deviation between the calculated curve and the experimental one is acceptable. In Fig. 9, the calculated enthalpies of formation of the compounds are compared with the experimental data from

Table 6
Summary of the evaluated thermodynamic parameters of the Ce-Si system[a].

Liquid: Model $(Ce, Si)_1$
${}^0L^{liquid}_{Ce,Si} = -276530 + 22.980 * T$ ${}^1L^{liquid}_{Ce,Si} = +24173 + 10.590 * T$
${}^2L^{liquid}_{Ce,Si} = 64797$
${}^3L^{liquid}_{Ce,Si} = -20314.7$
Ce_5Si_3: Model $(Ce)_5(Si)_3$
$CPCE5SI3 = -222.0388 * T * ln(T) - 0.024532 * T^2 + 318220 * T^{-1}$ ${}^0G^{Ce_5Si_3}_{Ce:Si} - 5 * {}^0H^{fcc}_{Ce} - 3 * {}^0H^{diamond}_{Si} = -585000 + 1124.2 * T + CPCE5SI3$
Ce_3Si_2: Model $(Ce)_3(Si)_2$
${}^0G^{Ce_3Si_2}_{Ce:Si} - 3 * {}^0G^{fcc}_{Ce} - 2 * {}^0G^{diamond}_{Si} = -331190 - 0.7160 * T$
Ce_5Si_4: Model $(Ce)_5(Si)_4$
${}^0G^{Ce_5Si_4}_{Ce:Si} - 5 * {}^0G^{fcc}_{Ce} - 4 * {}^0G^{diamond}_{Si} = -620870 - 12.167 * T$
CeSi: Model $(Ce)_1(Si)_1$
${}^0G^{CeSi}_{Ce:Si} - {}^0G^{fcc}_{Ce} - {}^0G^{diamond}_{Si} = -140160 - 5.7736 * T$ $CeSi_{1.34}$: Model $(Ce)_1(Si)_{1.34}$ ${}^0G^{CeSi_{1.34}}_{Ce:Si} - {}^0G^{fcc}_{Ce} - 1.34 * {}^0G^{diamond}_{Si} = -160000 - 7.7742 * T$
$CeSi_{1.68}$: Model $(Ce)_{.37}(Si)_{.63}$
${}^0G^{CeSi_{1.68}}_{Ce:Si} - 0.37 * {}^0G^{fcc}_{Ce} - 0.63 * {}^0G^{diamond}_{Si} = -66017 - 4.3563 * T$
$CeSi_2$: Model $(Ce)_1(Si)_2$
${}^0G^{CeSi_2}_{Ce:Si} - {}^0G^{fcc}_{Ce} - 2 * {}^0G^{diamond}_{Si} = -192040 - 8.4100 * T$

[a] In J mol-formula^{-1}; temperatures (T) in Kelvin. The Gibbs energies for the pure elements are from the SGTE compilation [25].

Table 7
Comparison of the invariant reactions among the calculations, experiments and evaluations.

Reaction	Type	Reference[a]	Type[b]	at.% Si			T (K)
$L \leftrightarrow (\gamma Ce) + Ce_5Si_3$	Eutectic	This work	▼	10.9	0	37.5	900.1
		2002Bul	▲	10.5	0	37.5	908
		2009Shu	▼	–	–	–	899
		2015Kim	▼	10.6	0	37.5	898.9
$L + Ce_3Si_2 \leftrightarrow Ce_5Si_3$	Peritectic	This work	▼	30.8	40	37.5	1536.2
		2002Bul	▲	–	–	–	1533
		2009Shu	▼	–	–	–	1550
		2015Kim	▼	32.6	40	37.5	1545.6
$L + Ce_5Si_4 \leftrightarrow Ce_3Si_2$	Peritectic	This work	▼	34.3	44.4	40	1610.3
		2002Bul	▲	–	–	–	1608 ± 5
		2009Shu	▼	–	–	–	1617
		2015Kim	▼	35.3	44.4	40	1600.6
$L + CeSi \leftrightarrow Ce_5Si_4$	Peritectic	This work	▼	41.9	50	44.4	1772.9
		2002Bul	▲	–	–	–	1773 ± 15
		2009Shu	▼	–	–	–	1790
		2015Kim	▼	42	50	44.4	1771.8
$L \leftrightarrow CeSi$	Congruent	This work	▼	50	50	–	1899.5
		2002Bul	▲	–	–	–	1903
		2009Shu	▼	–	–	–	1903
		2015Kim	▼	50	50	–	1913
$L \leftrightarrow CeSi + CeSi_{1.68}$	Eutectic	This work	▼	54.6	50	63	1853.6
		2002Bul	▲	55	–	–	1805.5 ± 12.5
		2004Sou	▲	–	–	–	1887
		2009Shu	▼	–	–	–	1848
		2015Kim	▼	54.5	50	62.5	1860.6
$CeSi + CeSi_{1.68} \leftrightarrow CeSi_{1.34}$	peritectoid	This work	▼	50	63	57.3	1469.1
		This work	▲	–	–	–	1469.0[c]
$L \leftrightarrow CeSi_{1.68}$	Congruent	This work	▼	63	63	–	1992.1
		2002Bul	▲	–	–	–	1993
		2004Sou	▲	–	–	–	1988
		2009Shu	▼	–	–	–	1998
		2015Kim	▼	62.5	62.5	–	2004.5
$L + CeSi_{1.68} \leftrightarrow CeSi_2$	Peritectic	This work	▼	72.1	63	66.7	1833.6
		2002Bul	▲	–	–	–	1833
		2009Shu	▼	–	–	–	1860
		2015Kim	▼	71	62.5	66.7	1859.2
$L \leftrightarrow CeSi_2 + (Si)$	Eutectic	This work	▼	83.2	66.7	100.0	1502.3
		This work	▲	–	–	–	1502.6[c]
		1913Vog	▲	81.5	–	–	1514
		1966Ben	▲	87	–	–	1463
		2002Bul	▲	82	–	–	1471 ± 2
		2009Shu	▼	–	–	–	1460
		2015Kim	▼	84.3	66.7	100	1465.6

[a] 2002Bul, [13]; 2009Shu, [3]; 2015Kim, [4]; 1913Vog, [16]; 1966Ben, [5].
[b] ▼stands for calculation, ▲ stands for experiment.
[c] DTA/DSC investigations were carried out in argon gas of P = 101325 Pa. Standard uncertainty is $u(P)$ = 5000 Pa, $u(T)$ = 2 K.

both the literature [6,7] and the present work (Table 8). It should be mentioned that the calculated value for Ce_5Si_3 is more negative than the experimental one, which is consistent with the previous calculation work [2–4]. Indeed, accepting the experimental enthalpy value exactly would result in the artificial decomposition in its stable range and the failure to reproduce the melting

Table 8
Calculated enthalpy of formation at 298 K along with the experimental data[a].

Ce_5Si_3	Ce_3Si_2	Ce_5Si_4	CeSi	$CeSi_{1.34}$	$CeSi_{1.68}$	$CeSi_2$	Method	Refs.
–	–	–	−71.1 ± 3.3	–	–	−60.5 ± 2.0	Calorimetry	[6]
−52.8 ± 0.6	–	–	–	–	–	–	Calorimetry	[7]
−61.0	−65.0	−69.0	−71.1	–	−65.0	−60.5	Calculated	[2]
−57.225	−60.9	−65.689	−70.5	–	−68.104	−62.733	Calculated	[3]
−63.44	−65.60	−67.78	−70.00	–	−63.75	−61.83	Calculated	[4]
–	–	–	–	–	−70 ± 6[c]	–	Calorimetry	This work
−64.310	−66.239	−68.985	−70.082	−68.376[b]	−66.017	−64.012	Calculated	This work

[a] In kJ mol-atoms.
[b] Fixed value.
[c] The uncertainty is expressed as standard uncertainty. The experiment was carried out in argon gas of 101325 ± 5000 Pa at 1373 ± 2 K.

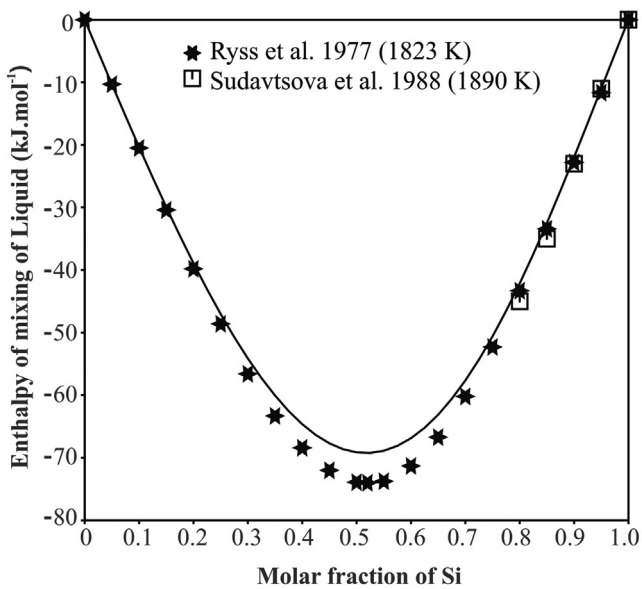

Fig. 8. Calculated enthalpies of mixing for the liquid phase of the Ce-Si system, compared with the experimental data [8,9,23].

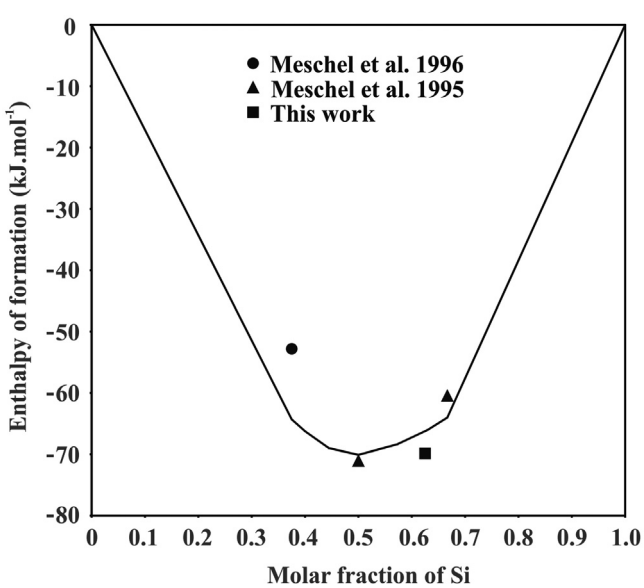

Fig. 9. Calculated enthalpies of formation of the compounds at 298.15 K along with the experimental data from both literature [6,7] and present work in the Ce-Si system.

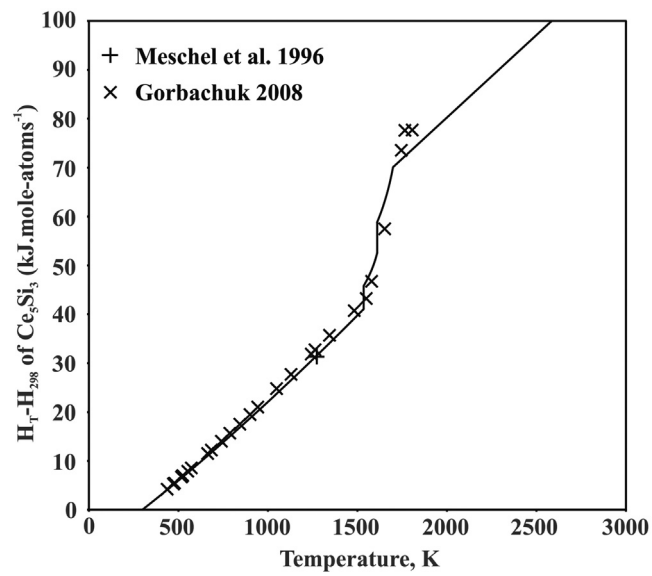

Fig. 10. Calculated H_T-H_{298K} of Ce_5Si_3 along with the measured data [7,10].

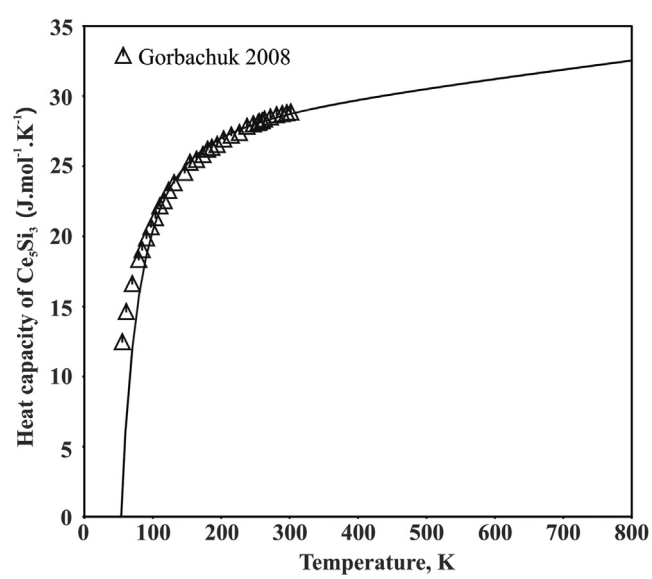

Fig. 11. Calculated heat capacity of Ce_5Si_3, compared with the experimental data [10].

point of Ce_5Si_3.The calculated enthalpy increments and heat capacities of Ce_5Si_3 using the presently obtained parameters are shown in Figs. 10 and 11, respectively. The comparison shows an excellent agreement between the calculations and the experimental data [7,10]. However, in order to reproduce the experimental heat

contents of the liquid part and the phase diagram, a more positive enthalpy of mixing of liquid and a more negative enthalpy of formation of Ce_5Si_3 are required, as seen in Figs. 8 and 9.

6. Conclusions

The Ce-Si system has been investigated by experiments and thermodynamics. Based on characterization using XRD, SEM, DTA and HTRC on the samples, some new characteristics were obtained. A phase with composition close to $CeSi_{1.34}$ was observed and the decomposition temperature of 1469 K was detected. The minimum Si content in $CeSi_{1.68}$ is determined to be 63.1 at.% at 1100 K. The eutectic reaction liquid $\leftrightarrow$ (Si) + $CeSi_2$ is observed to occur at 1502.6 K. The enthalpy of formation of $CeSi_{1.68}$ was measured to be -70 ± 6 kJ mol-atoms^{-1}.

All of the experimental phase diagram and thermodynamic data available for the Ce-Si system were critically evaluated. A substitutional solution model for the liquid phase was used and a set of self-consistent thermodynamic parameters for the Ce-Si system was obtained.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.tca.2016.11.003.

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